

A Simple Explanation of HOSVD and HOOI Using a $2 \times 2 \times 2$ Tensor

1. The Tensor

Consider the simple $2 \times 2 \times 2$ tensor

$$\mathcal{T}(:, :, 1) = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}, \quad \mathcal{T}(:, :, 2) = \begin{bmatrix} 5 & 6 \\ 7 & 8 \end{bmatrix}.$$

2. Mode Unfoldings

The three matricizations are:

$$T_{(1)} = \begin{bmatrix} 1 & 2 & 5 & 6 \\ 3 & 4 & 7 & 8 \end{bmatrix}, \quad T_{(2)} = \begin{bmatrix} 1 & 3 & 5 & 7 \\ 2 & 4 & 6 & 8 \end{bmatrix},$$
$$T_{(3)} = \begin{bmatrix} 1 & 3 & 2 & 4 \\ 5 & 7 & 6 & 8 \end{bmatrix}.$$

Each unfolding is 2×4 .

3. HOSVD (Higher-Order SVD)

HOSVD computes orthonormal factor matrices using the SVD of each unfolding:

$$T_{(n)} = U_n \Sigma_n V_n^T \quad (n = 1, 2, 3).$$

We keep all singular vectors (full rank $r_1 = r_2 = r_3 = 2$).

Thus we obtain:

$$U_1 \in \mathbb{R}^{2 \times 2}, \quad U_2 \in \mathbb{R}^{2 \times 2}, \quad U_3 \in \mathbb{R}^{2 \times 2}.$$

The core tensor is then computed by:

$$\mathcal{G} = \mathcal{T} \times_1 U_1^T \times_2 U_2^T \times_3 U_3^T.$$

HOSVD generalizes the matrix SVD:

$$\mathcal{T} = \mathcal{G} \times_1 U_1 \times_2 U_2 \times_3 U_3,$$

4. Truncated HOSVD

If we want a low-rank Tucker model, say target ranks $(1, 1, 1)$, we choose the leading singular vector from each unfolding:

$$U_1^{(1)}(:, 1), U_2^{(1)}(:, 1), U_3^{(1)}(:, 1).$$

This gives an initial guess for HOOI.

5. HOOI (Higher-Order Orthogonal Iteration)

HOOI improves the truncated HOSVD factors via alternating least squares.

At iteration k , update mode 1 by solving:

$$\max_{U_1^T U_1 = I} \left\| \mathcal{T} \times_2 \left(U_2^{(k)} \right)^T \times_3 \left(U_3^{(k)} \right)^T \right\|_F^2.$$

This is equivalent to:

$$\tilde{T}_{(1)} = T_{(1)}(U_3^{(k)} \otimes U_2^{(k)}),$$

$$U_1^{(k+1)} = \text{top left singular vectors of } \tilde{T}_{(1)}.$$

Similarly update:

$$U_2 \leftarrow \text{top singular vectors of } T_{(2)}(U_3 \otimes U_1),$$

$$U_3 \leftarrow \text{top singular vectors of } T_{(3)}(U_2 \otimes U_1).$$

Iterate until convergence:

Final core:

$$\mathcal{G} = \mathcal{T} \times_1 U_1^T \times_2 U_2^T \times_3 U_3^T.$$